L0A

Cell Lineages: Repeated Ancestral Paths Waste Memory

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Read it https://arxiv.org/abs/2104.00123 - Free to read (open-access preprint available)

Domain Biology & biomedical science

Abstract (quoted verbatim from the publication)

Reconstructing high-resolution single-cell phylogenetic lineages requires tracking cellular divisions across thousands of generations. Storing complete ancestral trajectories for every cell independently incurs exponential memory overhead due to repeated prefixes in early lineage branches. We formulate the lineage storage bottleneck as a prefix-redundancy problem and demonstrate that shared cellular ancestries require hierarchical prefix deduplication.

Demonstration paper excerpt prepared for Level 0 audience onboarding.

Notes: record the page or section for every claim you make about this paper.

Reading prompts

1. What problem does the paper identify regarding storing ancestral trajectories?
2. Why does independent cell storage waste memory on repeated ancestral prefixes?
3. What algorithmic structure could store shared prefixes once?

This card is a workshop wrapper prepared by the organisers. Only the abstract is quoted from the publication; read the paper itself through the link above before asserting any relationship.